

How do you like your dairy?

A review of how the world tastes milk products

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ABSTRACT

How you perceive and consume dairy is determined by various factors. One of these factors is culture. This paper looks at the relationship between dairy products and their country of origin based on textual descriptions. To do this, Wikipedia pages are used to find dairy products and their origin. Moreover, descriptors are defined to find possible relationships between product and country. The results are validated using human judgement and other sources, and show...

Code available at: <https://github.com/dewi-elisa/dairyWorld>.

1 INTRODUCTION

Dairy consumption varies across cultures. For example, ethnicity affects the liking of yogurt [10]. Moreover, the same dairy products can be perceived as differing in healthiness across cultures [4]. In addition, country characteristics can influence which dairy products are consumed [9]. Therefore, studying the relationship between people and dairy can clarify the broader human relationship with food and deepen the understanding of cultural differences.

Research question. What is the relationship between dairy products and the country of origin?

- For each dairy product, where do they originate from?
- How is the dairy product consumed?
- Is there a relationship between the dairy product and the country of origin?

2 RELATED WORK

Dairy perception and consumption are strongly shaped by culture. At the same time, the history and geography of dairy farming influence how dairy products are produced and valued, and the country of origin of a product can influence consumers. This section goes into more depth on these topics and further discusses food computing and the current research.

2.1 Dairy and culture

A number of studies show that people in different countries perceive and like dairy products in different ways. For example, work on yogurt with different degrees of granularity [10] found that Chinese and Danish consumers responded differently to the presence or absence of particles. Moreover, Johansen et al. [4] studied young Norwegian, Danish, and Californian consumers and showed that all three groups have common motives for consuming calorie-reduced dairy products (e.g., fat content, perceived healthiness and taste), but their relative importance differs across countries. In addition,

research on Australian and Chinese consumers [9] reports cross-country differences in the liking of pasteurized milk versus long shelf life milk and in how shelf life labeling affects liking. Lastly, research with a trained panel on flavor lexicons for cheese [1] showed that the panels from New Zealand, Ireland, and the United States used different words to describe the same cheeses, and tended to group cheeses by country of origin.

Beyond sensory perception, cultural and historical factors also influence how dairy is produced, valued, and consumed. For example, in India most milk comes from the buffalo rather than cows because cows have a sacred status in Hinduism. Moreover, in the United States, people are increasingly shifting towards plant-based milks [20]. Lastly, dairy farming can be of great importance to mountain communities [17], family farming cultures [3], and animal welfare [16]. This is further emphasized by Valenze [19], who researched the different religious meanings of milk and the different functions throughout the ages. Thus, dairy is not only a food product, but also has cultural, religious, and socio-economic significance.

2.2 Country-of-origin labeling

Country-of-origin (COO) research has shown that the country where a product is made can influence how consumers evaluate it. This suggests that links between dairy products and countries can shape perceptions and choices. Knight and Calantone [5] propose a model of how COO information is processed cognitively and show that COO effects are complex and cultural context plays an important role in purchase decisions. Moreover, research on the Canadian market [13] found that COO labeling affects the purchase choice and willingness to pay for dairy products. Finally, Yang et al. [22] illustrate how consumer behavior is influenced through COO information after a major food-related scandal.

2.3 Food computing

Food computing and computational gastronomy are emerging research areas that use computational methods to study food-related topics. Surveys and overviews in this area [2, 12] describe how food data (e.g., recipes, images, social media posts) can be analyzed with tools from data science and artificial intelligence to address a wide range of questions about flavor, health, and culinary culture. Several studies focus on culinary cultures and food-country relations. For example, Laufer et al. [8] mine cultural relations between different language communities in Europe by analyzing how food cultures are described on Wikipedia pages across different language editions. Moreover, palette varieties around the world are investigated [15] and are found to have strong geographical and cultural similarities between countries. Furthermore, a large visual question answering dataset with text-image pairs is made identifying dish names and their origins [21]. Lastly, two topic modelling

approaches are applied to recipes [7, 11] to discover more about the human relationship with food.

2.4 Current research

Previous research has shown that culture influences how dairy products are perceived and consumed, and that the country of origin of a product can affect how consumers evaluate it. Moreover, work on food computing has demonstrated that large-scale online data and computational methods can be used to study food-related topics, including culinary cultures and food-country relations. However, there is little work that combines these perspectives to study how dairy products and countries are linked in textual descriptions. Therefore, this paper uses text mining on Wikipedia articles to investigate how dairy products are associated with their countries of origin.

3 APPROACH

3.1 Data

For this project, Wikipedia pages of dairy products and countries are used. All pages are linked in a Wikipedia page with a list of dairy products¹. Only the products with all information present (product name, product URL, origin(s), origin URL(s)) are used, resulting in a total of 60 dairy products out of 130. All data is scraped on 5 January 2026 using the BeautifulSoup library in Python² and further processed using pandas [14].

3.2 Method

The linked Wikipedia page provides a list of dairy products and origins, answering the first subquestion. To answer the second subquestion, descriptors are defined to identify processing and preparation methods and serving contexts of dairy products (see Tables 1a and 1b). These descriptors are then used to search on the dairy product pages for paragraphs mentioning one of the keywords, extracting the descriptor, matched keywords and sentence in which the keyword occurred. Each dairy product is then matched with one of the 5 descriptors from each category by assigning the one with the most keyword hits. If multiple descriptors had the highest amount of hits, both descriptors are assigned. The same is done for animal sources of dairy products (see Table 1c), trying to answer the third subquestion. To fully answer the third subquestion, the results from the descriptors are visualized in a world map.

3.3 Validation

For the first subquestion, the results are validated by randomly picking 6 dairy products (10%) and checking their origins with other sources.

For the second subquestion, 10% of the extracted paragraphs are randomly selected and manually checked for correctness.

Lastly, for the third subquestion, the country pages are searched for paragraphs containing the dairy product, and the dairy product pages for paragraphs that mention the country of origin. Those paragraphs are extracted and used to validate the results.

¹https://en.wikipedia.org/wiki/List_of_dairy_products

²<https://pypi.org/project/beautifulsoup4/>

Table 1: Descriptors and their keywords used to identify processing and preparation methods, serving contexts, and animal sources of dairy products.

(a) Processing and preparation methods	
Descriptor	Keywords
Fermented	fermented, fermentation, culture, lactic, probiotic, kefir, yogurt, yoghurt
Aged	aged, ripened, ripening, mature, affinage
Fresh	fresh, unripened
Smoked	smoked, smoky
Heat treated/cooked	cooked, baked, heated, melted, pasteurized, pasteurised
(b) Serving context	
Descriptor	Keywords
Breakfast	breakfast, morning, brunch, granola, cereal, oatmeal, porridge
Bread	bread, toast, sandwich, spread, schmear, bagel, cracker, crostini, canape, canapés
Dessert	dessert, sweet, cake, pastry, pastries, pudding, custard, ice cream, gelato, whipped, frosting
Cooking	cooking, sauce, pasta, casserole, gratin
Snack	snack, appetizer, appetiser, cheese board, charcuterie, platter, tapas
(c) Animal source	
Descriptor	Keywords
Cow	cow, bovine, jersey, holstein
Goat	goat, caprine, chèvre, chevre
Sheep	sheep, ewe, ovine
Buffalo	buffalo, bufala
Plant-based	plant-based, plant based, vegan, non-dairy, non dairy, dairy-free, dairy free, oat, soy milk, soya milk, almond milk, coconut milk, cashew milk, rice milk, pea milk

4 RESULTS

4.1 Product origins

Figure 1 shows the origins of the 60 dairy products from Wikipedia. Darker shades indicate a higher number of products originated from that country. An interactive version is available on GitHub³. The interactive version shows on hover which specific products originated from that country. Appendix B describes the mapping procedure.

From Figure 1 it can be seen that most countries have just a few products, while Denmark and Norway have 4, Turkey has 6, and India has 7. Most dairy products are from Europe or Asia, while very few come from Africa and Oceania. Note that some products have multiple origins, so totals reflect product-country links rather than unique products.

³https://dewi-elisa.github.io/dairyWorld/results/world_map.html

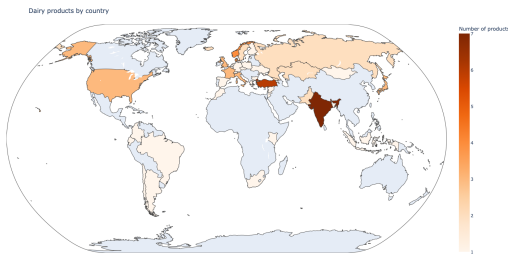


Figure 1: Origins of dairy products

4.2 Consumption methods

Figure 2 summarizes the descriptors assigned to each of the dairy products after matching the descriptor keywords from Table 1 against the text of the corresponding Wikipedia product pages. Since multiple descriptors can be assigned to the same product, the percentages within each plot do not necessarily sum to 100%.

For processing and preparation (Figure 2a), *fermentation*-related keywords were matched most frequently across products. Moreover, for serving context (Figure 2b), most products matched with a *dessert*-related keyword. Lastly, for animal source (Figure 2c) the most frequent outcome was *none*, indicating that many products did not mention a specific animal source in the text.

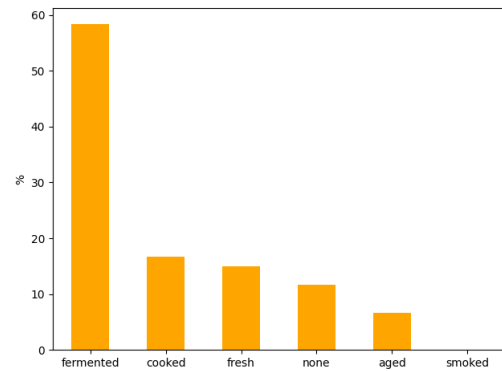
4.3 Product-country relations

Figure 3 visualizes the geographic distribution of each descriptor by showing, for every country, the percentage of products associated with that country that were assigned the selected descriptor. Note that products can be linked to multiple countries of origin and can be assigned multiple descriptors. Therefore, a single product may contribute to multiple countries and to multiple descriptor maps. Interactive versions of Figure 3 are available on GitHub⁴.

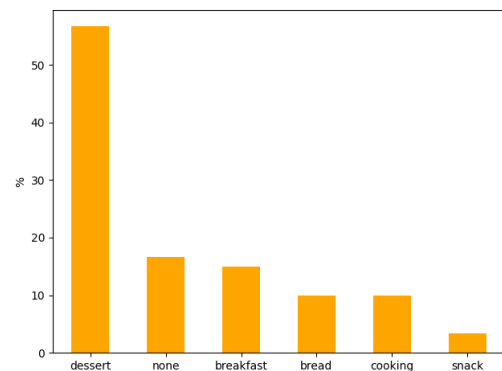
The descriptor maps in Figure 3 suggest that descriptors differ in how geographically widespread they are within the dataset. For processing and preparation, *fermented* is the most frequently assigned descriptor (35/60 products; 58.3%) and appears across many countries, whereas *cooked* is less widespread geographically and is most visible in European countries (10/60 products; 16.7%).

For serving context, *dessert* is the dominant descriptor (34/60; 56.7%) and is geographically widespread, while *bread* appears mainly in the United States and Europe (6/60; 10.0%). Moreover, *cooking* appears more strongly in countries around the Mediterranean (6/60; 10.0%).

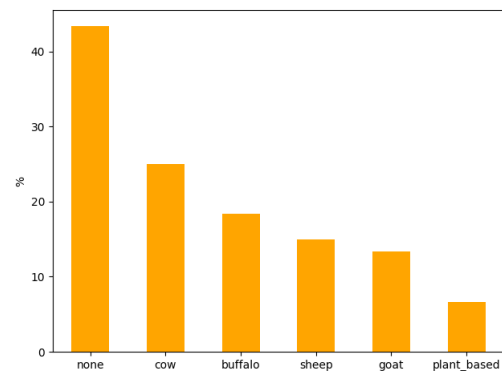
Lastly, for animal-source descriptors, only 34/60 products (56.7%) contained an explicit animal-source match in the text, meaning that *none* is the most frequent outcome (26/60; 43.3%). Among matched animal descriptors, *cow* is the most common overall (15/60; 25.0%), while *buffalo* (11/60; 18.3%) and *sheep* (9/60; 15.0%) show more geographically concentrated patterns (*buffalo* matches are mainly visible in South America and South Asia, whereas *sheep* matches are most visible in Russia and several Mediterranean countries).



(a) Processing and preparation descriptors



(b) Serving descriptors



(c) Animal source descriptors

Figure 2: Descriptor occurrences across dairy products. Note that multiple labels can apply to one product, and therefore the percentages do not sum to 100%.

⁴https://dewi-elisa.github.io/dairyWorld/results/descriptor_map_dropdown.html

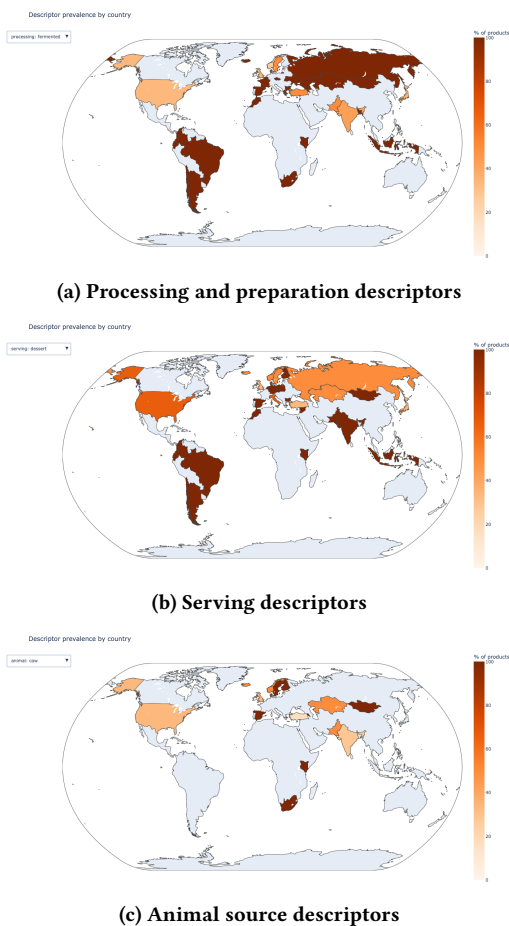


Figure 3: Descriptor prevalence of each descriptor by country.

4.4 Validation

5 DISCUSSION

5.1 Validation

* What the validation implies for trustworthiness * How it affects interpretation of your main findings (do results hold, where to be cautious).

5.2 Limitations

The current research only used 60 dairy products, while the Wikipedia page consisted of 130 products. Many products were removed because not all metadata fields were complete (e.g., missing origin or missing link to a dedicated Wikipedia page). Some of this information could have been added manually or with more extensive scraping (e.g., because the origin was mentioned in the description column rather than the origin column of the dairy product table). Since the number of products is limited, this can result in unstable country-level patterns, where countries with only one or two products can appear to have very high (or very low) descriptor prevalence in Figure 3.

Furthermore, this research only takes into account textual descriptions from Wikipedia. The data contained ambiguous regional terms, such as Scandinavia, Vikings or Central Asia. Therefore, some compromises were made by mapping ambiguous regional/cultural references to modern countries (see also Appendix B). This may introduce misclassification and affect country-level counts and descriptor prevalence. In addition, food is not only described but also experienced through smell, taste, and texture. The absence of sensory context may bias the descriptors toward what is typically written rather than what is perceived.

Moreover, since the data is scraped from the internet, there might be a bias present to more well-documented countries and regions. Origins that are less well-documented might have additional dairy products which were not found on Wikipedia. This likely contributes to the geographic imbalance visible in Figure 1. Therefore, the map should be interpreted as Wikipedia coverage of products with complete metadata, rather than a complete representation of global dairy diversity or production.

Lastly, it is important to note that the country of origin does not necessarily mean that the product is also consumed there. It might well be that a product originated in country X, travelled to country Y, and became forgotten in country X while country Y is still eating the product. Therefore, not all assumptions about the country of origin and how a product is consumed are warranted.

5.3 Future work

Future work could focus on adding more data sources. When adding additional data sources, data from other modalities could also be used to make up for the absence of sensory context. Additionally, more products could be added, either by additional data sources or by adding missing information to the dairy products that were removed due to missing data.

Moreover, a more in-depth analysis could be made. This could, for example, be done by choosing different descriptors. The current descriptors were handpicked, but another choice could be to make data-driven ones, for example by learning descriptor groups from text embeddings or topic models.

Future work could also incorporate multilingual or non-Wikipedia sources (e.g., regional food databases, cookbooks, or protected designation lists) to reduce English- and Wikipedia-specific bias. Finally, origin could be compared to consumption by incorporating recipe datasets or market data.

6 CONCLUSION

In order to investigate the relationships between dairy products and their country of origin, Wikipedia pages were scraped and used to extract relationships. The results show... While there are some apparent limitations, they can easily be overcome by improving on this work.

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A WORK REPORT

In order to find literature, I mainly looked in an online university library. I used queries such as 'milk', 'dairy', 'country of origin dairy', and 'food computing'.

During the programming, I mainly looked at documentation and examples online on websites such as <https://www.geeksforgeeks.org> and <https://www.w3schools.com>. One problem I encountered was the Robot Policy of Wikipedia, https://wikitech.wikimedia.org/wiki/Robot_policy#REST_API_rules, which prevented me from scraping the Wikipedia pages. I solved this by adding a user-agent to my requests. For determining the descriptors from Table 1, I asked ChatGPT-5.2 for keywords. The exact prompt was: *Hi, can you think of descriptors and keywords that say more about dairy? I would like a set of descriptors about: * the processing and preparation of dairy (e.g., fermented, smoked, fresh) * serving context (e.g., breakfast, bread) * the animal source (e.g., goat, cow, sheep, plant) Give about 5 descriptors per category.*

Country	ISO3
USSR	Russia
Khan garh	Pakistan
Scotland	United Kingdom
Vikings	United Kingdom
Central Asia	Kazakhstan
West Sumatra	Indonesia
Scandinavia	Sweden, Norway, Denmark
Alps	Switzerland
Andhra Pradesh	India
Caucasus	Georgia
Punjab	India, Pakistan
Central and Eastern Europe	Poland
Mannheim	Germany
Minoru Shirota	Japan

Table 2: Manual mapping of non-country origins to countries for ISO3 conversion

B WORLD MAP

In order to be able to make Figure 1, ISO3 mapping was needed for each country of origin. The country converter package [18] was used for this task. However, not all countries were countries: some were cities, regions, former countries, or cultures. Therefore, a manual mapping was made for these cases to a country. The manual mapping to that country is shown in Table 2. For the cities and regions, the country where they are located was chosen. For the former countries, the country that is most closely related to them was chosen. For the Viking culture, the United Kingdom was chosen as the Wikipedia page with dairy product mentions Scotland. When a part of a continent was mentioned, a country central in that part was chosen. Together with product and country names, this was then put in a choropleth map using Plotly [6].